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EDS ACTIVITY 1

TOPIC: FOR LOOP AND ONE EXAMPLE





FOR LOOP IN PYTHON

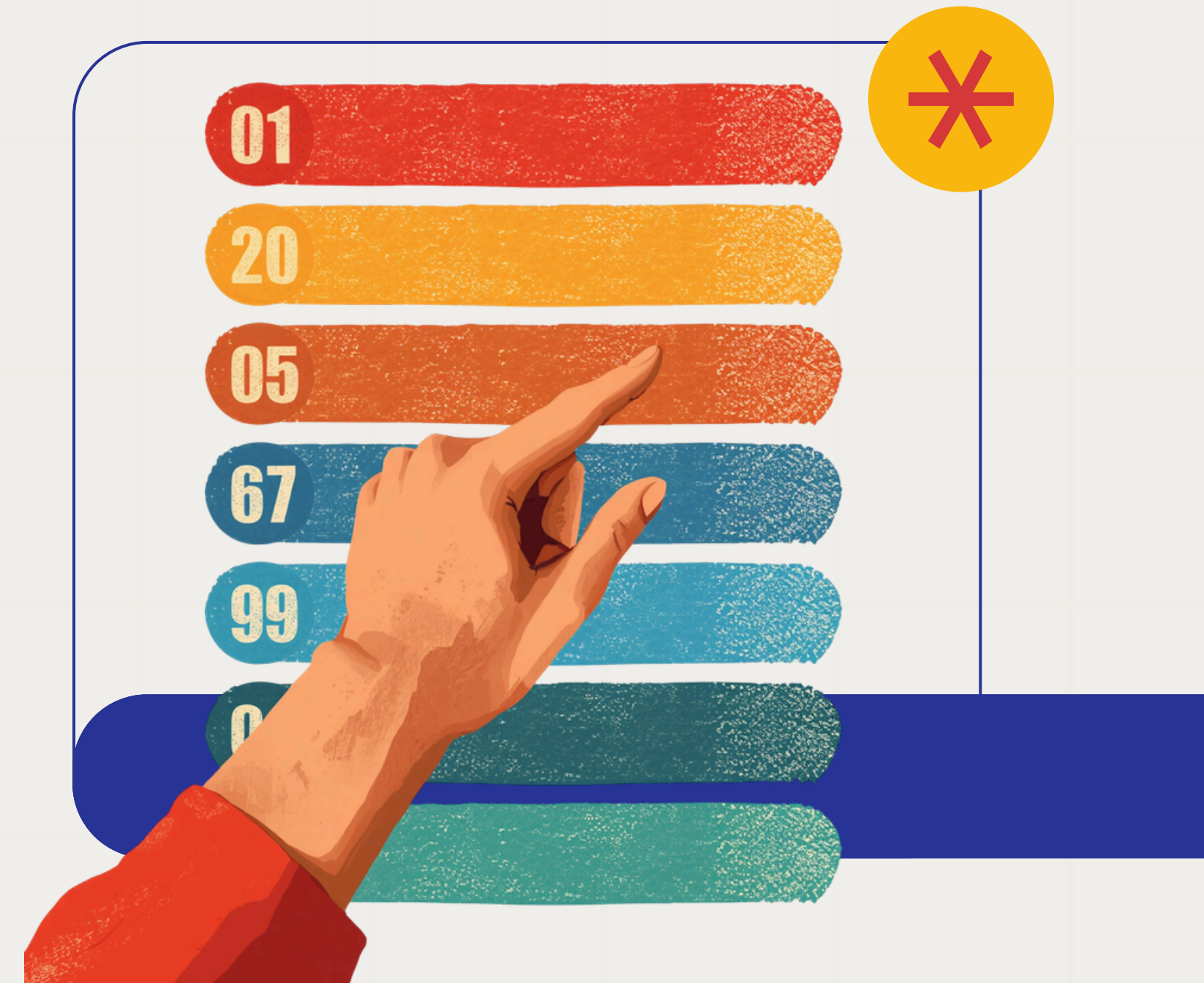
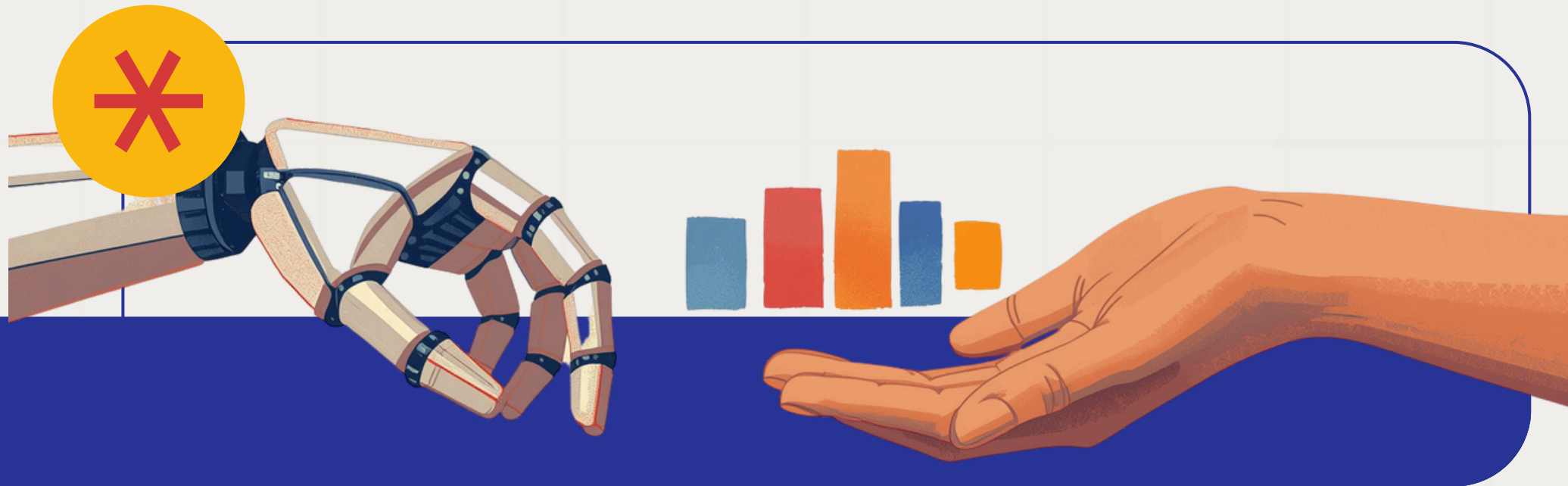




WHAT IS A LOOP?

A loop is a programming construct that repeats a block of code multiple times. Loops enable efficient execution of repetitive tasks, reduce code redundancy, and make programs cleaner and easier to maintain.

EX; FOR LOOP,
WHILE LOOP



WHAT IS A FOR LOOP?

A for loop iterates over a sequence such as lists, strings, or range objects. It automatically processes each element one by one, eliminating the need for manual indexing. This makes code cleaner and more efficient.

FOR LOOP SYNTAX

for i in range(n):
print(i)

- i = loop variable (changes each iteration)
- range(n) = iterable (sequence of numbers)
- print(i) = code block (indented, runs each loop)

for variable_name in iterable_object:

Code block to be executed in each iteration
This block must be indented





EXAMPLE:

MULTIPLICATION TABLE

Use a for loop with the range function to iterate from 1 up to (but not including) 11

```
for i in range(1, 11):
```

```
    product = 5 * i
```

```
    print(f"5 * {i:2} = {product:3}")
```





CONCLUSION

For loops are fundamental to programming and data science. They enable efficient data processing, automation of repetitive tasks, and form the foundation for complex data science workflows and analysis.

